

AN2DL - Second Homework Report

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1 Introduction

In this second challenge, we face an *semantic segmentation* problem on a mars surface dataset. Contrary to the first challenge, we put more focus on the models **architectures and losses**, because we already learnt to implement a consistent **augmentation** pipeline from last time.

Our approach to this challenge was to spend less time on the basic models and make use of the literature and our successive endeavours to search for more intelligent models which would adapt to this problem.

Since we couldn't make use of pretrained models, **hyperparameters** tuning and training efficacy was paramount.

2 Problem Analysis

The dataset is initially composed of about 2615 pairs of grayscale images and their corresponding label mask. These labels are divided into 5 classes with a great imbalance, especially for the "Big Rocks" class which represents less than 1% of label pixels. Finding a loss which takes this imbalance into account became a central point of our work.

No missing labels were identified; however, visualizing samples with same labels, we discovered and dropped 110 outliers (images labelled as aliens).



Figure 1: Alien outlier label and sample

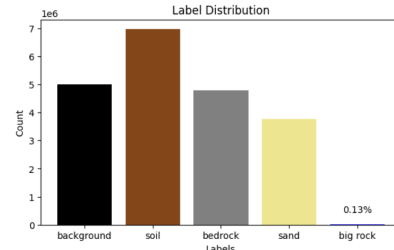


Figure 2: Class distribution after cleaning

We performed PCA on the image features extracted by a VGG16 in order to visualise the classes in their latent representation (notebook attached).

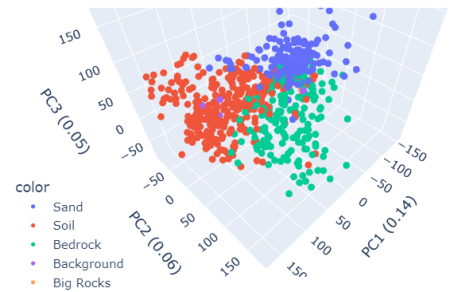


Figure 3: Images features from VGG16 in the first principle plane of the PCA

This representation shows how Sand, Bedrock and Soil are much easier to identify than Background and Big rocks which have less distinctive features or are less represented. A weighted loss could be the solution to this issue.

3 Method

3.1 Baseline

From both the raw train and test dataset images, we applied a common preprocessing step, normalizing to a $[0, 1]$ range for a more effective training, added a new axis for compatibility with tensor processing, and brightness (mean normalised to 0.5) and contrast adjustments, for improved feature differentiation among similar pixel areas.

We then split the dataset into train and validation sets, reserving 300 samples for validation and 2205 for training, and applied data augmentation to improve our training, by building a dataset via `tf.dataset.Dataset`: flips, Gaussian noise (common in digital images), random zoom (to occult wrong background labels on the extreme borders) are our 3 transformations.

Our model chosen as baseline is the same presented in the Segmentation lab of the course, thus a very simple yet efficient architecture for the problem in question.

3.2 Simple Double UNet

The Simple Double UNet employs two sequential UNets to improve segmentation. The first UNet generates a global segmentation mask, which is concatenated with the input image and fed into the second UNet for refined, localized predictions. This architecture leverages skip connections and encoder-decoder blocks for feature preservation and enhanced segmentation accuracy.

To balance class distribution, we used from `scikit-learn` the library `compute_class_weight` function. The weights were assigned inversely proportional to the frequency of each class in the dataset. After computing the weights, the weight for the "Big Rock" class was divided by 10 or more, as its computed value was excessively high due to its extreme rarity. Without this adjustment, the model would have over-prioritized the class at the expense of others. The weights were then incorporated into the

loss function for the training.

3.3 Double UNet with enhanced bottleneck

This model enhances the Simple Double UNet by incorporating an Enhanced Bottleneck in each UNet. The bottleneck uses parallel dilated convolutions for multi-scale feature extraction and a simplified Squeeze-and-Excitation mechanism to prioritize relevant features. This refinement improves segmentation precision, particularly in complex tasks.

About augmentation, more images were augmented for training samples with Big Rocks, as an attempt to further balance the distribution.

4 Experiments

4.1 Self-Attention and Adaptive Weighted Loss

From the 2024 paper of MarsSeg model [1], we tried to implement a **self-attention** [2] mechanism combined with an adaptive loss. Attention would allow the model to detect dependencies between different areas across the image, which could be especially useful for the background class. The adaptive loss is a combination of **Focal and Dice Loss** and updates the class weights at each epoch based on the mean IoU of each class over the epoch. The classes that are worse classified get weighted more for the following. We could also give different weights to each of the losses. However, this method didn't do better than our double UNet.

4.2 Advanced Double UNet

We introduced gated skip connections to dynamically learn gating weights, filtering encoder features before concatenation with decoder features. While promising in theory, this approach did not improve the mean validation IoU and added architectural complexity, so it was excluded from the final model.

Additionally, a custom normalization layer was implemented to address dataset distribution but failed to provide significant benefits. Similarly, replacing `UpSampling2D` with `Conv2DTranspose` for upscaling did not yield expected results due to checkerboard artifacts and less precise feature reconstruction. As none of these methods enhanced

Table 1: Our testing progresses. Best results are highlighted in **bold**.

Model	Parameters	Val IoU	Test IoU
Baseline	1.8M	0.4232	0.4199
Simple Double UNet	3.7M	0.4081	0.4452
Enhanced Double UNet	5.7M	0.5199	0.5219
Advanced Double UNet	6.3M	0.4465	0.4447
Self-attention UNet and Adaptive Loss (Dice+Focal)	9.8M	0.4812	0.4262

performance, they were omitted from the final model.

5 Results

The performance of each model on the validation and test sets is summarized in Table 1. The Baseline model achieved a test IoU of 0.4199, serving as the starting point of our experiments. The Simple Double UNet showed a slight improvement with a test IoU of 0.4452, indicating that the sequential application of UNets helped refine predictions. The Enhanced Double UNet outperformed the other models with a test IoU of 0.5219, validating the effectiveness of the enhanced bottleneck for multi-scale feature extraction and attention mechanisms for better segmentation precision.

Although the Self-Attention and Adaptive Weighted Loss model seemed promising, it did not outperform the Enhanced Double UNet, achieving lower validation and test IoU values. The adaptive weighting approach, while theoretically beneficial for addressing class imbalances, likely required further tuning for this specific dataset.

6 Discussion

The results highlight the critical role of architectural enhancements in addressing semantic segmentation challenges on imbalanced datasets. The Enhanced Double UNet achieved notable gains over the baseline by leveraging dilated convolutions and Squeeze-and-Excitation mechanisms in the bottleneck, enabling improved multi-scale feature extraction and feature prioritization.

The use of a self-attention mechanism underscored the potential for capturing long-range dependencies. Its underperformance likely reflects sub-optimal hyperparameter tuning or dataset limita-

tions, suggesting further exploration with advanced attention-based architectures or larger datasets.

The adaptive loss mechanism showed promise but lacked consistent improvements, possibly due to instability in dynamically updating weights during training. However, combining Focal and Dice Loss partially addressed class imbalance, and fine-tuning loss weight parameters could yield better results.

7 Conclusions

This report examined strategies for semantic segmentation of Mars surface images, emphasizing architectural innovations and loss functions to address class imbalance. The Enhanced Double UNet emerged as the most effective model, achieving a test IoU of 0.5219, significantly surpassing the baseline. This underscores the value of multi-scale feature extraction and fine detail attention in complex segmentation tasks.

Although the self-attention mechanism and adaptive loss function did not outperform the Enhanced Double UNet, they provided insights for future improvements. Combining these approaches with larger datasets or pretraining could unlock further performance gains.

In summary, the Enhanced Double UNet presents a promising method for challenging semantic segmentation tasks. Refining loss functions and exploring attention-based architectures offer exciting directions for future research.

References

- [1] Junbo Li, Keyan Chen, Gengju Tian, Lu Li, and Zhenwei Shi. MarsSeg: Mars Surface Semantic Segmentation with Multi-level Extractor and Connector. *arXiv*, 2024.

- [2] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y Hammerla, Bernhard Kainz, Ben Glocker, and Daniel Rueckert. Attention U-Net: Learning Where to Look for the Pancreas. *arXiv*, 2018.